

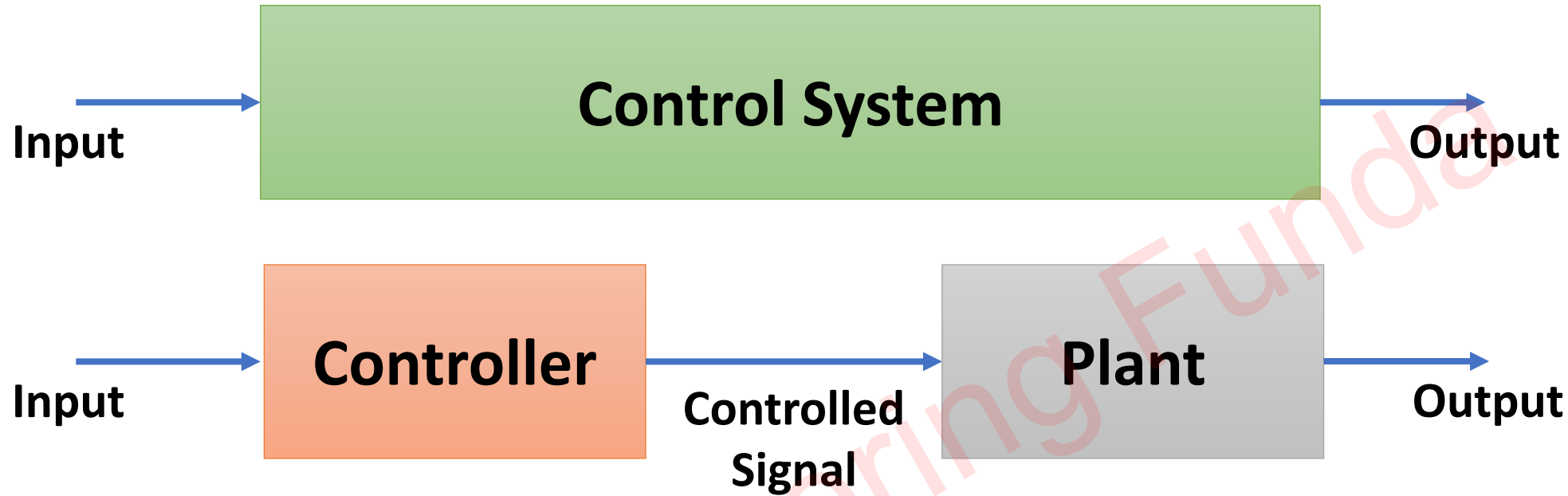
Open Loop System

Outlines

- Block Diagram of Open Loop System
- Applications of Open Loop System
- Advantages of Open Loop System
- Disadvantages of Open Loop System

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Block Diagram of Open Loop System



- Open Loop System doesn't have any Feedback.
- Controller Generates Controlled Signal with the information of Input.
- In Open Loop System, Control Action is independent of Output of plant.
- Controlled Signal is also referred as manipulated Signal.

Applications of Open Loop System

- Volume of Audio System



- Door Lock System



- Hair Dryer



- Washing Machine



Advantages of Open Loop System

- Simplicity {They are simple in their Layout}
- Cost Effective {As Simple in Structure}
- It is convenient to use when output is difficult to measure, as output of controller does not effected by output of the system.
- Mostly Stable {As there is no Feedback}
- Fast Response

Disadvantages of Open Loop System

- Inaccurate {As Controller doesn't Controlled signal with the use of Output}
- Unreliable {System Response doesn't depends on Output}
- Manual Offset Correct {Automatic Error Correction is not possible}
- Sensitivity to Disturbances
- Not Suitable for Complex System

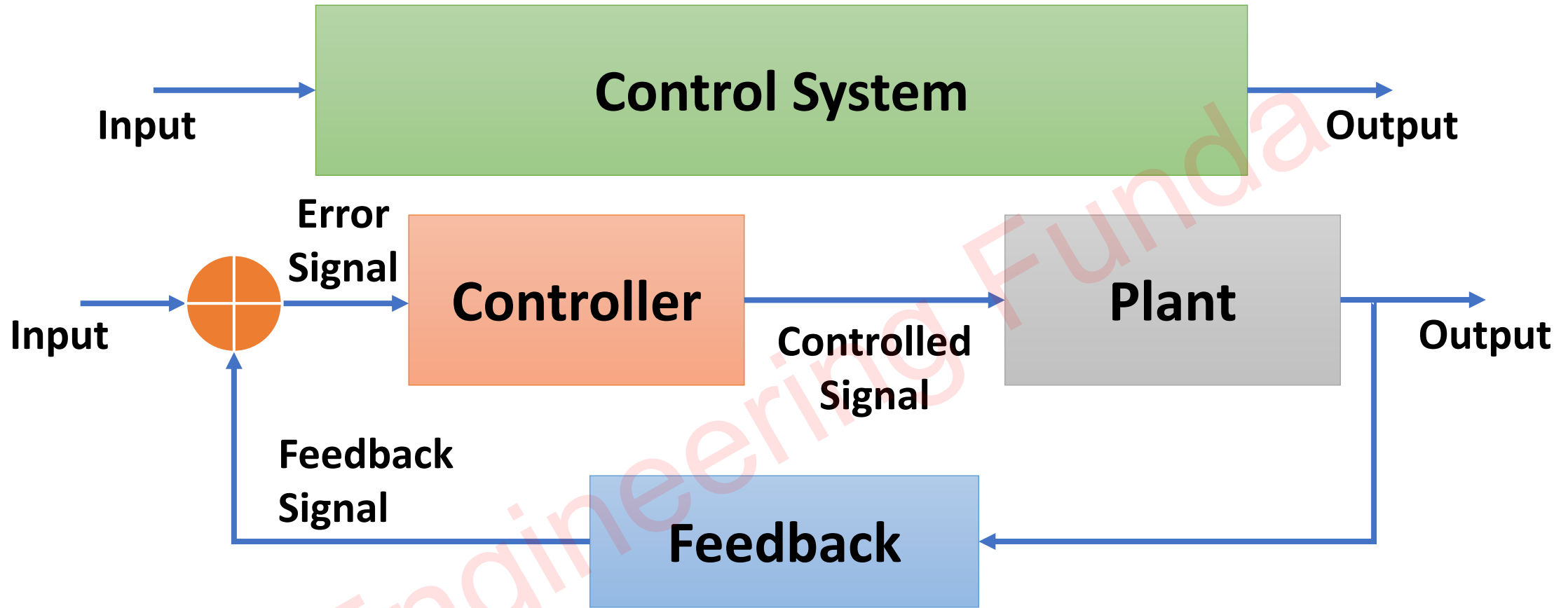
Closed Loop System

Outlines

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Block Diagram of Closed Loop System



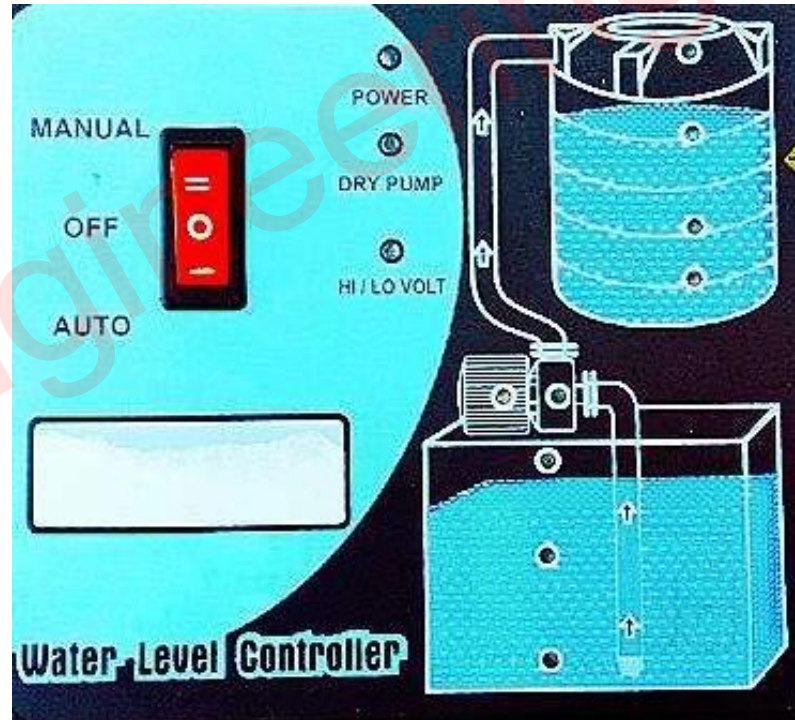
- Closed Loop System has Feedback From Output.
- Controller Generates Controlled Signal with the information of Error Signal.
- In the Closed Loop System, Control Action depends on the Input and Output of the plant.
- Due to Feedback, the Output Response of the System is more accurate.

Applications of Closed Loop System

- Air Conditioner



- Water Level Controller

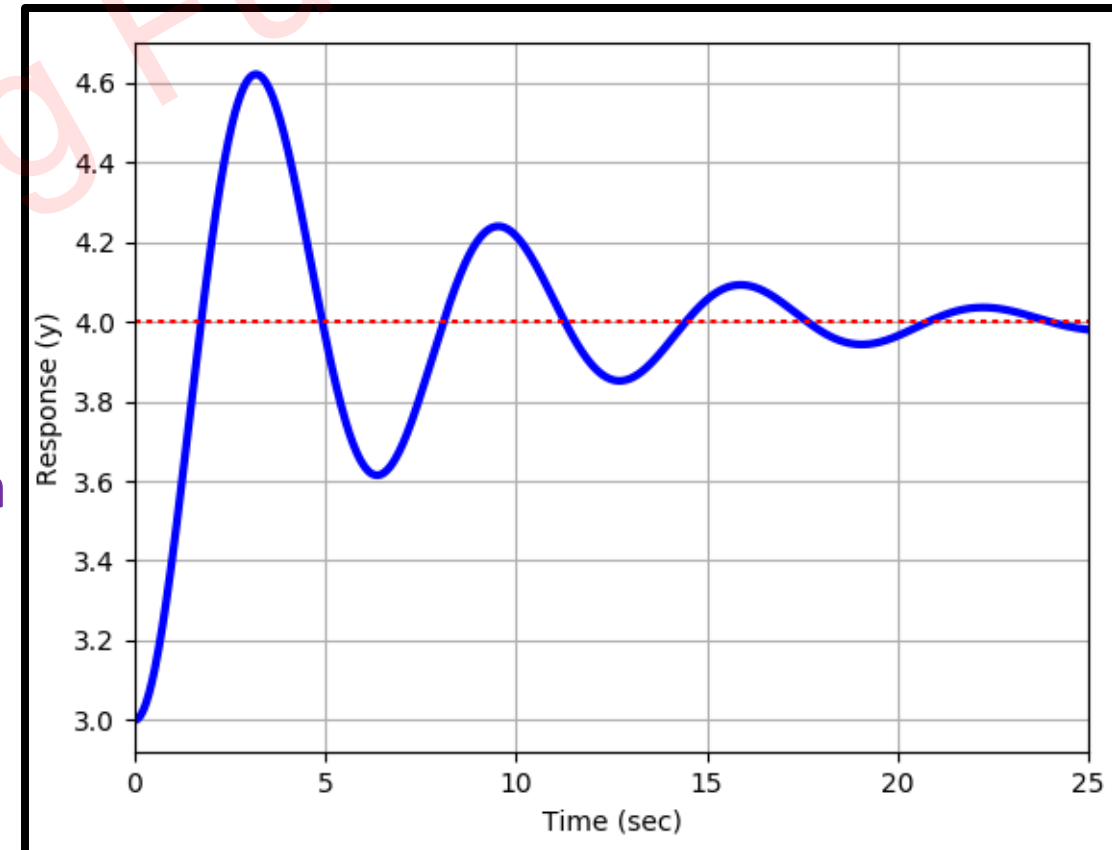


Advantages of Closed Loop System

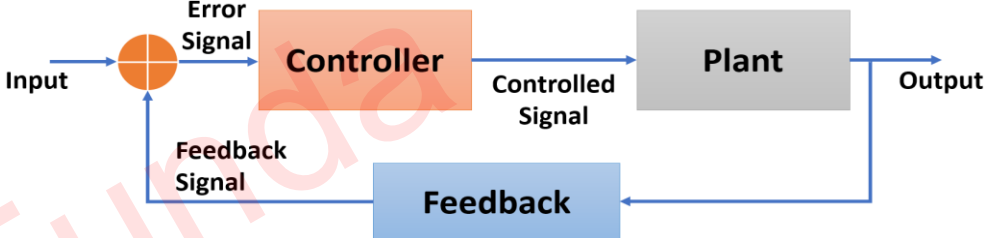
- Accurate even in the presence of Non-linearities
- Accurate even if an Error arises
- Large Bandwidth
- Facilitates Automation
- Less Sensitive to Disturbances
- Less Affected by Noise
- Adoptability

Disadvantages of Closed Loop System

- Costly and Complex
- Feedback may make a system unstable
- Feedback reduces the overall gain of the system
- Requires more Maintenance
- Slower Response
- Feedback leads to oscillatory response



Comparison of Open Loop System and Closed Loop System

Parameters	Open Loop System	Closed Loop System
1. Block Diagram		
2. Feedback	Not available	Available
3. Error Detector	Not available	Available
4. Accuracy	Less	More
5. Sensitivity	High	Less
6. Bandwidth	Less	More
7. Stability	Mostly Stable	Due to Feedback it may get unstable
8. Cost	Less	More
9. Implementation	Easy	Complex
10. Oscillation in response	Rare	Due to Feedback it may have oscillation
11. Maintenance	Easy and Cheap	Complex and Costly
12. Speed	Fast	Slow

Feedbacks in Control System

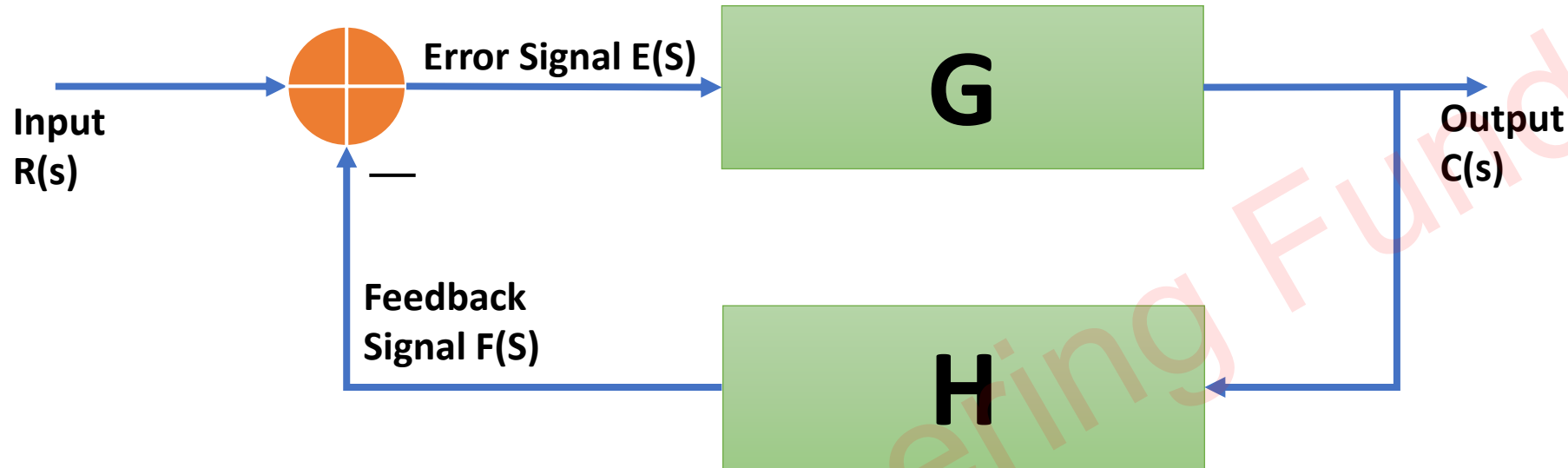
Types of Feedback

- Negative Feedback
- Positive Feedback

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Negative Feedback

- If we Subtract the Feedback Signal concerning Input, it will be Negative Feedback.



Applications

- ❑ Stabilize gain
- ❑ Amplifier
- ❑ In ADC/DAC

❖ Here Feedback $F(S)$ is

$$\Rightarrow F(S) = HC(S)$$

❖ Output $C(S)$ is

$$\Rightarrow C(S) = GE(S)$$

❖ Here error signal $E(S)$ will be

$$\Rightarrow E(S) = R(S) - F(S)$$

$$\Rightarrow C(S)/G = R(S) - HC(S)$$

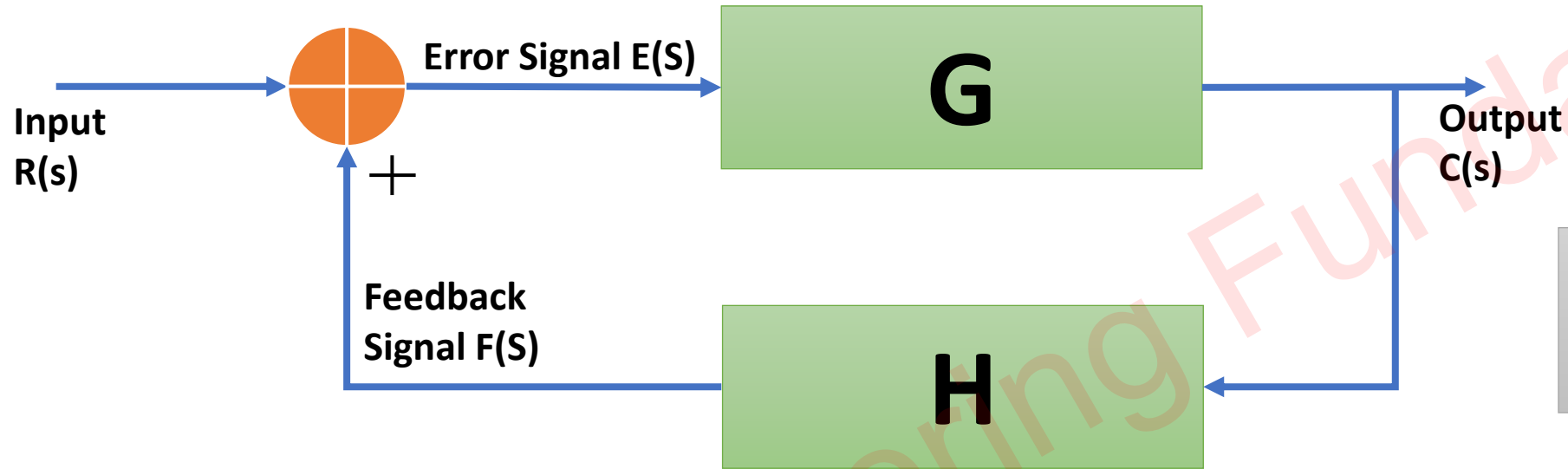
$$\Rightarrow C(S)(1+GH) = GR(S)$$

❖ So Transfer function is

$$\Rightarrow \frac{C(s)}{R(S)} = \frac{G}{1 + GH}$$

Positive Feedback

- If we Add the Feedback Signal concerning Input, it will be Positive Feedback.



Applications

- ❑ Oscillator
- ❑ In Timing Circuits

❖ Here Feedback $F(S)$ is

$$\Rightarrow F(S) = HC(S)$$

❖ Output $C(S)$ is

$$\Rightarrow C(S) = GE(S)$$

❖ Here error signal $E(S)$ will be

$$\Rightarrow E(S) = R(S) + F(S)$$

$$\Rightarrow C(S)/G = R(S) + HC(S)$$

$$\Rightarrow C(S)(1 - GH) = GR(S)$$

❖ So Transfer function is

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{G}{1 - GH}$$

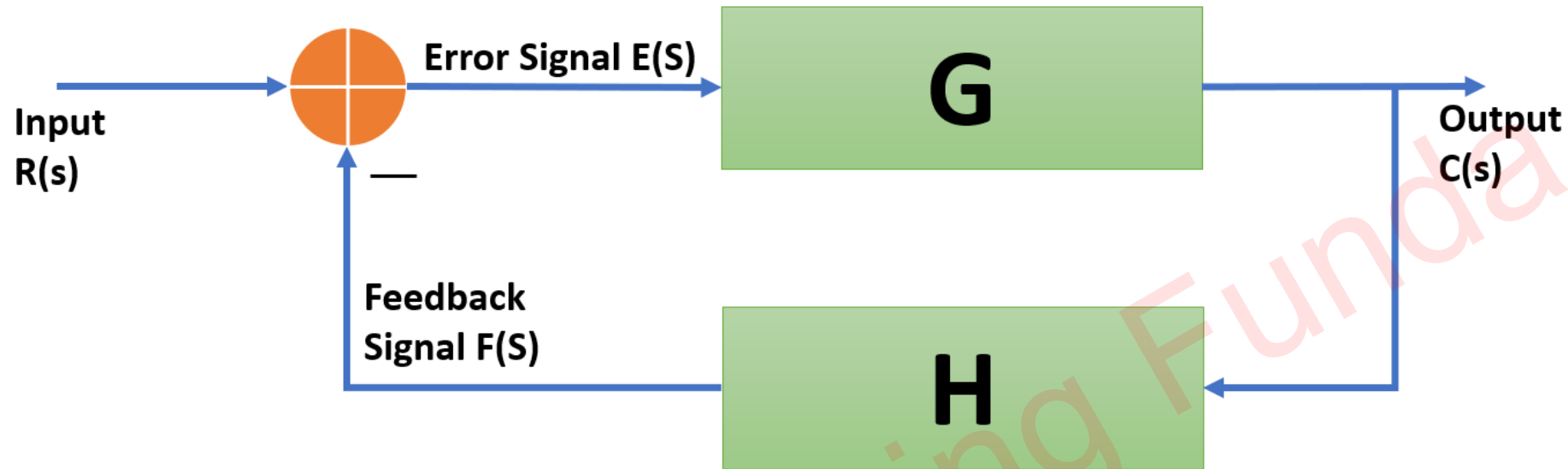
Effects of Feedback

Three Major Effects of Feedback

- Effect of Feedback on Gain
- Effect of Feedback on Sensitivity
- Effect of Feedback on Stability

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Effect of Feedback on Gain



- So Transfer function with feedback is

$$\Rightarrow T(s) = \frac{C(s)}{R(s)} = \frac{G}{1 + GH}$$

- Gain of system without feedback is G

- So effect can be studied from TF

- ☐ If H is increasing
- ☐ Then $(1+GH)$ will get increase
- ☐ So $T(s)$ will decrease

- ☐ If H is Decreasing
- ☐ Then $(1+GH)$ will get Decrease
- ☐ So $T(s)$ will Increase

- ☐ So by Changing H , we can change TF.

Effect of Feedback on Sensitivity

- So TF with -Ve feedback is

$$\Rightarrow T(s) = \frac{C(s)}{R(s)} = \frac{G}{1 + GH}$$

- Sensitivity is defined by

$$\Rightarrow S = \frac{\% \text{ Change in } T(s)}{\% \text{ Change in } G(s)}$$

$$\Rightarrow S = \frac{(dT/T \times 100)}{(dG/G \times 100)}$$

$$\Rightarrow S = \frac{dT}{dG} \times \frac{G}{T}$$

- From Transfer function

$$\Rightarrow \frac{G}{T} = 1 + GH$$

- Differentiate T with respect to G

$$\Rightarrow \frac{dT}{dG} = \frac{d}{dG} \left[\frac{G}{1 + GH} \right]$$

$$\Rightarrow \frac{dT}{dG} = \frac{(1 + GH) \left(\frac{dG}{dG} \right) - (G) \left(\frac{d(1 + GH)}{dG} \right)}{(1 + GH)^2}$$

$$\Rightarrow \frac{dT}{dG} = \frac{(1 + GH)(1) - (G)(H)}{(1 + GH)^2}$$

$$\Rightarrow \frac{dT}{dG} = \frac{1}{(1 + GH)^2}$$

- So, Sensitivity will be

$$\Rightarrow S = \frac{dT}{dG} \times \frac{G}{T} = \frac{1}{1 + GH}$$

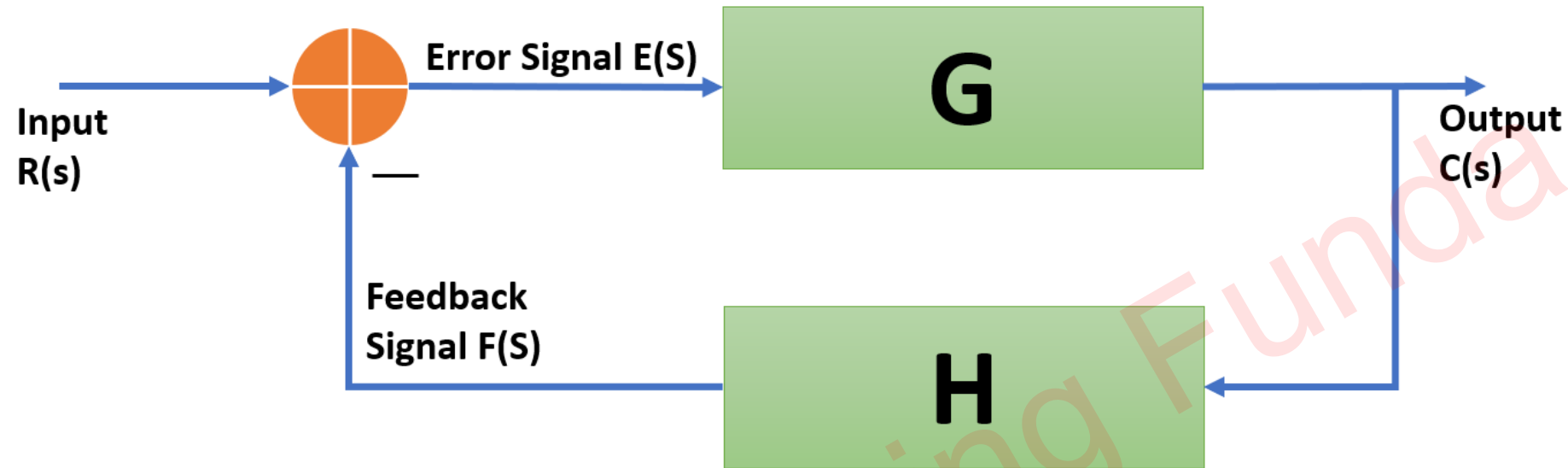
□ If H is increasing

□ Then (1+GH) will get increase

□ So S will decrease and Visa versa

□ So Sensitivity and feedback gain is reciprocal to each other.

Effect of Feedback on Stability



- So Transfer function with feedback is

$$\Rightarrow T(s) = \frac{C(s)}{R(s)} = \frac{G}{1 + GH}$$

- Gain of system without feedback is G

- So if $H = -1/G$, then Transfer function will get infinity, so feedback may make system unstable.

❑ So Due to feedback, stability of system decreases.